

$$(AA^+)^T = AA^+,$$

$$(A^+A)^T = A^+A.$$

Problems

28-1 Tridiagonal systems of linear equations

Consider the tridiagonal matrix

$$A = \begin{pmatrix} 1 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{pmatrix}.$$

- a. Find an LU decomposition of A .
- b. Solve the equation $Ax = (1 \ 1 \ 1 \ 1 \ 1)^T$ by using forward and back substitution.
- c. Find the inverse of A .
- d. Show how to solve the equation $Ax = b$ for any $n \times n$ symmetric positive-definite, tridiagonal matrix A and any n -vector b in $O(n)$ time by performing an LU decomposition. Argue that any method based on forming A^{-1} is asymptotically more expensive in the worst case.
- e. Show how to solve the equation $Ax = b$ for any $n \times n$ nonsingular, tridiagonal matrix A and any n -vector b in $O(n)$ time by performing an LUP decomposition.

28-2 Splines

A practical method for interpolating a set of points with a curve is to use **cubic splines**. You are given a set $\{(x_i, y_i) : i = 0, 1, \dots, n\}$ of $n + 1$ point-value pairs, where $x_0 < x_1 < \dots < x_n$. Your goal is to fit a piecewise-cubic curve (spline) $f(x)$ to the points. That is, the curve $f(x)$ is made up of n cubic polynomials $f_i(x) = a_i + b_i x + c_i x^2 + d_i x^3$ for $i =$

$0, 1, \dots, n-1$, where if x falls in the range $x_i \leq x \leq x_{i+1}$, then the value of the curve is given by $f(x) = f_i(x - x_i)$. The points x_i at which the cubic polynomials are “pasted” together are called **knots**. For simplicity, assume that $x_i = i$ for $i = 0, 1, \dots, n$.

To ensure continuity of $f(x)$, require that

$$f(x_i) = f_i(0) = y_i,$$

$$f(x_{i+1}) = f_i(1) = y_{i+1}$$

for $i = 0, 1, \dots, n-1$. To ensure that $f(x)$ is sufficiently smooth, also require the first derivative to be continuous at each knot:

$$f'(x_{i+1}) = f'_i(1) = f'_{i+1}(0)$$

for $i = 0, 1, \dots, n-2$.

a. Suppose that for $i = 0, 1, \dots, n$, in addition to the point-value pairs $\{(x_i, y_i)\}$, you are also given the first derivative $D_i = f'(x_i)$ at each knot. Express each coefficient a_i, b_i, c_i , and d_i in terms of the values y_i, y_{i+1}, D_i , and D_{i+1} . (Remember that $x_i = i$.) How quickly can you compute the $4n$ coefficients from the point-value pairs and first derivatives?

The question remains of how to choose the first derivatives of $f(x)$ at the knots. One method is to require the second derivatives to be continuous at the knots:

$$f''(x_{i+1}) = f''_i(1) = f''_{i+1}(0)$$

for $i = 0, 1, \dots, n-2$. At the first and last knots, assume that $f''(x_0) = f''_0(0) = 0$ and $f''(x_n) = f''_{n-1}(1) = 0$. These assumptions make $f(x)$ a **natural** cubic spline.

b. Use the continuity constraints on the second derivative to show that for $i = 1, 2, \dots, n-1$,

$$D_{i-1} + 4D_i + D_{i+1} = 3(y_{i+1} - y_{i-1}). \quad (28.23)$$

c. Show that

$$2D_0 + D_1 = 3(y_1 - y_0) , \quad (28.24)$$

$$D_{n-1} + 2D_n = 3(y_n - y_{n-1}) . \quad (28.25)$$

- d.* Rewrite equations (28.23)–(28.25) as a matrix equation involving the vector $D = (D_0 \ D_1 \ D_2 \ \cdots \ D_n)^T$ of unknowns. What attributes does the matrix in your equation have?
- e.* Argue that a natural cubic spline can interpolate a set of $n + 1$ point-value pairs in $O(n)$ time (see Problem 28-1).
- f.* Show how to determine a natural cubic spline that interpolates a set of $n + 1$ points (x_i, y_i) satisfying $x_0 < x_1 < \cdots < x_n$, even when x_i is not necessarily equal to i . What matrix equation must your method solve, and how quickly does your algorithm run?

Chapter notes

Many excellent texts describe numerical and scientific computation in much greater detail than we have room for here. The following are especially readable: George and Liu [180], Golub and Van Loan [192], Press, Teukolsky, Vetterling, and Flannery [365, 366], and Strang [422, 423].

Golub and Van Loan [192] discuss numerical stability. They show why $\det(A)$ is not necessarily a good indicator of the stability of a matrix A , proposing instead to use $\|A\|_\infty \|A^{-1}\|_\infty$, where $\|A\|_\infty = \max \{\sum_{j=1}^n |a_{ij}| : 1 \leq i \leq n\}$. They also address the question of how to compute this value without actually computing A^{-1} .

Gaussian elimination, upon which the LU and LUP decompositions are based, was the first systematic method for solving linear systems of equations. It was also one of the earliest numerical algorithms. Although it was known earlier, its discovery is commonly attributed to C. F. Gauss (1777–1855). In his famous paper [424], Strassen showed that an $n \times n$ matrix can be inverted in $O(n^{\lg 7})$ time. Winograd [460] originally proved that matrix multiplication is no harder than matrix inversion, and the converse is due to Aho, Hopcroft, and Ullman [5].

Another important matrix decomposition is the *singular value decomposition*, or *SVD*. The SVD factors an $m \times n$ matrix A into $A = Q_1 \Sigma Q_2^T$, where Σ is an $m \times n$ matrix with nonzero values only on the diagonal, Q_1 is $m \times m$ with mutually orthonormal columns, and Q_2 is $n \times n$, also with mutually orthonormal columns. Two vectors are *orthonormal* if their inner product is 0 and each vector has a norm of 1. The books by Strang [422, 423] and Golub and Van Loan [192] contain good treatments of the SVD.

Strang [423] has an excellent presentation of symmetric positive-definite matrices and of linear algebra in general.

¹ The year in which *Introduction to Algorithms* was first published.